

AMRUTHANAMAPURE

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in PCB design, circuit development, and embedded systems. Background includes production and quality testing exposure at Avalon Technologies and an embedded systems internship building a Li-ion Battery Management System on STM32. Skilled in VLSI design basics, Verilog, embedded systems, and Python. Seeking an entry-level PCB Design, Embedded Hardware, or BMS/Power Electronics Engineering role.

EXPERIENCE

Production Department, Testing Line | Avalon Technologies

Apr 2026 – Aug 2026

- Worked in the testing line within the Production department, supporting quality and functional testing of electronic assemblies.
- Gained hands-on exposure to industrial production workflows, test procedures, and hardware quality checks.

Embedded Systems Intern | Vision Astraa EV Academy

Feb 2026 – May 2026

- Designed and developed a 5S Lithium-ion Battery Management System (BMS) using STM32 to monitor cell voltages, current, and temperature in real time.
- Collaborated with a team on embedded firmware development and hardware debugging for EV battery applications.

PROJECTS

Battery Management System (BMS) for Li-ion Battery Pack

- Designed a 5S Li-ion BMS using STM32F103C8T6 to monitor cell voltages, current, and temperature in real time (developed during Vision Astraa internship – core skill differentiator).

Portable Oscilloscope using Arduino & OLED

- Built a low-cost oscilloscope for real-time waveform visualization using ADC sampling and OLED display with adjustable scaling.

All-in-One Electronic Component Tester

- Built an Arduino-based tester to identify and measure resistors, capacitors, diodes, and transistors using ADC-based analysis with OLED display output.

AgriGuard: Smart IoT Farming & Security System

- Developed an Arduino-based system for automated irrigation and field security using soil moisture, PIR, and temperature sensors with ESP8266 IoT monitoring.

SKILLS

- PCB Designing and Fabrication
- Circuit Design
- VLSI Design Basics
- Embedded Systems
- Python
- Verilog

EDUCATION

Bachelor of Engineering (ECE) | City Engineering College, Bengaluru

2023 – 2026

CGPA: 7.70

Diploma in Electronics & Communication Engineering | Govt. Polytechnic, Athani, Belagavi

2020 – 2023

CGPA: 8.3

CERTIFICATIONS&TRAINING

- IoT&ItsApplications – ZeelCodeLabs
- Robotics Technician Vocational Training – GTTC (Jan 2023 – Feb 2023)
- EmbeddedSystems Workshop – ARMCortex

LANGUAGES

Kannada, English, Hindi, Marathi